

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CSE440 Capstone Project

<Project Progress Report>

ProHealth AI Assistant

Intelligent Hospital Portal & Bio_ClinicalBERT Triage System

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1. PROJECT OVERVIEW

1.1. Executive Summary

ProHealth AI Assistant is an end-to-end intelligent hospital management portal and clinical triage referral system developed to eliminate patient misdirection, optimize hospital resource utilization, and accelerate emergency routing. Powered by **Bio_ClinicalBERT** (emilyalsentzer/Bio_ClinicalBERT, pre-trained on MIMIC-III & PubMed) and fine-tuned on the benchmark **MTSamples (Medical Transcriptions) Dataset** (mtsamples.csv), the system analyzes unstructured multilingual complaints (Pure Bengali, phonetic Banglish, and English) and maps them across **13 core medical specialties** with confidence probabilities. Simultaneously, a deterministic rule-based urgency triage evaluator identifies critical red-flag triggers across 3 levels (Emergency, Urgent, Routine). The application integrates Google Identity Services OAuth, interactive doctor booking, persistent database tracking, and ReportLab PDF referral tickets, and is fully deployed live on **Render.com**.

1.2. Project Stakeholders

Role / Field	Details
Project Due Date	Final Evaluation / Capstone Submission 2026
Project Core Team (PCT)	Tasmiad Hasan (Lead AI/ML Engineer & Full-Stack Developer)
Other Stakeholders	Course Supervisors, Evaluation Board, Clinical Consultants

1.3. Milestones and Deliverables

ID	Milestone Name	Target Date	Actual Date	Status	Comments
M1	MTSamples Dataset Annotation	10 Jul 2026	10 Jul 2026	Achieved	13 clinical specialties mapped
M2	Bio_ClinicalBERT Fine-Tuning	20 Jul 2026	20 Jul 2026	Achieved	77% accuracy / 1.00 F1 key
M3	FastAPI Backend & Triage API	28 Jul 2026	28 Jul 2026	Achieved	Prediction & PDF endpoints
M4	Web Portal & Bilingual I18N	05 Aug 2026	05 Aug 2026	Achieved	Bangla/English toggle & UI
M5	Google Auth & User Dashboard	12 Aug 2026	12 Aug 2026	Achieved	GIS OAuth & persistence
M6	PDF Referral & Cloud Deploy	16 Aug 2026	16 Aug 2026	Achieved	Live on Render.com

1.4. Project Plan (per Work Progress)

Work Package	Planned Start	Planned End	Actual Start	Actual End	Progress (%)
WP1: Data Engineering	01 Jul 2026	10 Jul 2026	01 Jul 2026	10 Jul 2026	100%
WP2: Model Training (Colab)	11 Jul 2026	22 Jul 2026	11 Jul 2026	20 Jul 2026	100%
WP3: Backend REST API	23 Jul 2026	30 Jul 2026	23 Jul 2026	28 Jul 2026	100%
WP4: Web Portal & I18N	31 Jul 2026	08 Aug 2026	31 Jul 2026	05 Aug 2026	100%
WP5: Auth, Dashboard & PDF	09 Aug 2026	15 Aug 2026	09 Aug 2026	14 Aug 2026	100%
WP6: Cloud Deploy & QA	15 Aug 2026	17 Aug 2026	15 Aug 2026	16 Aug 2026	100%

2. PROJECT DETAILS

2.1. Scope Changes

ID	Category	Title & Description	Status	Size	Prio	Approved By	Date
SC 1	Technical	Google OAuth Migration: Switched from Clerk to Google Identity Services (GIS)	Done	3	5	PCT / Sup	15 Aug
SC 2	New Req	Vector PDF Generator: Server-side ReportLab streaming referral tickets	Done	4	4	PCT	14 Aug
SC 3	Technical	Bilingual I18N & Banglish: Dynamic Bangla/English UI & phonetic normalizer	Done	3	5	PCT	08 Aug
SC 4	Technical	Render Cloud Deployment: CPU wheel optimization for free cloud tier	Done	2	4	PCT	16 Aug

Footnotes: Size/Priority scale: 5=Very High, 4=High, 3=Medium, 2=Low, 1=Very Low.

2.2. Major Risks and Actions Taken

ID	Risk Name	Description	Status	L	I	RL	Strategy	Action Details
R1	Cold Start Delay	BERT loading from disk on first web request causing >10s latency	Closed	4	4	16	Reduce	Cached model in RAM at startup
R2	Cloud Size Quota	GPU PyTorch wheels (>2GB) exceeding free cloud memory limits	Closed	5	4	20	Avoid	Used CPU wheel index
R3	Auth Origin Block	Google OAuth blocking requests from unlisted live cloud URLs	Closed	4	5	20	Reduce	Whitelisted Render domain in GCP
R4	Medical Triage Error	Potential misclassification of emergency cases	Closed	2	5	10	Reduce	Added rule-based red-flag layer

Risk Level $RL = L \times I$ (Scale 1-25). Strategy: Avoid, Transfer, Reduce, Accept.

2.3. Major Issues and Actions Taken

ID	Title	Description & Status	Action Taken	Urg	Imp	Size	Owner
I1	Blank PDF Bug	Canvas capture rendering black screen [Resolved]	Switched to ReportLab vector generation	5	5	3	Tasmiad
I2	JSON Persistence	Missing module import skipping booking saves [Resolved]	Imported json and added error handling	5	4	1	Tasmiad
I3	Navbar Stacking	Buttons breaking into 2 rows on laptops [Resolved]	Enforced flexbox nowrap & height:42px	4	3	2	Tasmiad
I4	I18N Desync	Modal text remaining Bangla on English toggle [Resolved]	Populated I18N dict & DOM hooks	4	4	2	Tasmiad

2.4. Other On-Going and Planned Actions

Actions	Due Date	Who & Comments
Interactive Chatbot Triage Assistant: Multi-turn symptom questioning agent	Q4 2026	Tasmiad Hasan — Handling underspecified patient complaints
Hospital EHR / FHIR Protocol Integration: Direct database sync	Q1 2027	Tasmiad Hasan — Integration with hospital management software
Automated SMS & WhatsApp Booking Delivery: Instant SMS notifications	Q1 2027	Tasmiad Hasan — Twilio gateway integration for booking tickets

2.5. Achievements

Project Highlights / Achievements	Comments
State-of-the-Art Clinical Transformer (Bio_ClinicalBERT)	Fine-tuned BERT model on Google Colab, achieving 77.0% overall accuracy and 1.00 F1-score on key departments (Ophthalmology, Urology, Gastroenterology).
End-to-End Hospital Management Web Portal	Built responsive hospital website with 13 doctor directories, facilities explorer, voice recognition, and instant AI triage referral.
Seamless Google Identity OAuth & Dashboard	Implemented 1-tap Google login, appointment persistence in data/appointments.json, and active patient management.
100% Reliable Vector PDF Referral Generator	Implemented server-side ReportLab vector PDF generator embedding patient name, Google ID, department, and doctor recommendations.
Live Production Cloud Deployment	Successfully deployed and hosted live on Render.com (https://prohealth-ai-assistant.onrender.com/) with automated CI/CD from GitHub.